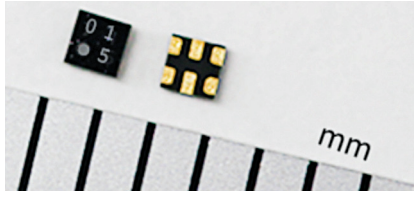
events reduces external raw data transfer, lowering latency and power consumption for real-time systems. It targets machine vision, robotics, physical AI, depth sensing, spectroscopy, scientific/medical imaging, quantum tech, and industrial automation.
Singular Photonics
https://singularphotonics.com

Ultra-low signal booster for space-constrained uses

Nisshinbo Micro Devices Inc. is set to launch the NT1741, an ultra-low-current receiver (Rx) booster for 2.4GHz wireless communication applications



such as Bluetooth Low Energy (BLE), Thread, and Zigbee. Operating across the 2400MHz to 2500MHz band, the device delivers a 13dB typical gain while drawing an industry-leading 1.5mA in LNA mode and just 1µA in bypass mode to prolong battery life. The NT1741 integrates all input and output matching circuits on-chip, requiring only a single external power-supply bypass capacitor. Housed in a tiny 1.0mm × 1.0mm EPFFP-6-FB package, this design minimises board real estate for space-constrained applications like hearing aids, AI glasses, and wireless headphones.
Nisshinbo Micro Devices Inc.
https://www.nisshinbo-microdevices.co.jp

CAN XL transceiver for greater payloads and data rates

Texas Instruments (TI) has introduced the TCAN6062, a controller area network extended data-field length (CAN XL) transceiver supporting payloads of up to 2048 bytes per frame and data rates of up to 20Mbps. Optimised for motion control, sensor feedback, and diagnostics in humanoid robotics and human-machine interface (HMI) systems, the device is backward-compatible with CAN FD and CAN SIC networks. Its integrated SIC function reduces signal ringing by up to 80 per



cent to simplify high-node network validation. Notably, the transceiver supports TCP/IP tunneling, allowing engineers to route Ethernet traffic over mixed-network architectures while utilising broad I/O voltage protection up to ±58V. The SIC function reduces ringing by up to 80 per cent in complex networks, simplifying validation in high-node architectures.
Texas Instruments (TI)
https://www.ti.com

PMIC to simplify EV traction inverter design

Infineon Technologies AG has introduced the OPTIREG TLE9744QK, an all-in-one power management IC for



electric and hybrid vehicle traction inverters that integrates power-supply functions, resolver excitation, and a safety engine. By generating the resolver signal internally and supplying multiple power rails, it eliminates external monitoring circuits and can reduce the replaced components' PCB footprint by up to 70 per cent. Built to ISO 26262 standards for ASIL-D compliance, the device supports existing MCU platforms and offers off-board sensor supply flexibility.
Infineon Technologies AG
https://www.infineon.com

CPU that can reduce the number of servers required

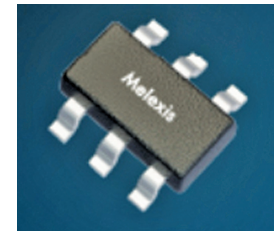
Fujitsu Limited's FUJITSU-MONAKA CPU has a maximum operating frequency of 3.8GHz and a memory transfer speed of 8800MT/s. The CPU delivers twice the AI inference throughput of other CPUs while using less power for equivalent workloads,



potentially reducing the number of servers required. It uses a 3D-stacked architecture, with a 2nm process for processor cores and a 5nm process for cache and input/output sections. Fujitsu combines this architecture with low-voltage operation technology to improve performance and power efficiency. It supports confidential computing, using hardware encryption to protect data and applications in memory for AI workloads in multi-tenant cloud environments.
Fujitsu Limited
https://global.fujitsu/en-global

3D sensor for cost-sensitive high-volume applications

Melexis has introduced the MLX90391, a 3D magnetometer designed for cost-sensitive, high-volume applications such as smart utility meters, smart home devices, and white goods. Featuring a magnetic measurement range of up to 500mT, the IC detects external magnetic fields used to interfere with



smart meter operations while reducing PCB redesign and manufacturer qualification requirements. The component supports contactless position sensing across rotary and linear controls, smart knobs, liquid-level sensing, electronic locks, handheld tool triggers, gaming controls, and e-bike gear position sensing. Other applications include replacing mechanical switches, sliders, and potentiometers in human-machine interface (HMI) functions where contactless sensing can reduce wear and tolerance-related issues.
Melexis
https://www.melexis.com